

Kuldeep Singh

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iOS Engineer · Ex-Walmart iOS Developer III · Swift / SwiftUI / On-Device AI

SUMMARY

iOS engineer with 3 years of production Swift inside 5 years of software engineering, across enterprise mobile and on-device AI. At Walmart (iOS Developer III), shipped navigation, performance, and analytics work on the flagship US shopping app (**100M+ downloads**). Founder of YouxAI, a Chrome-extension product running LLM inference fully on-device via WebGPU. Recently root-caused a GPU compiler bug corrupting on-device model output on NVIDIA/Windows and shipped the fix upstream; the platform bug is an open P2 Chromium issue.

CORE SKILLS

iOS / Mobile: Swift, SwiftUI, UIKit, Objective-C, Combine, Swift Concurrency, MVVM-C, Coordinator Pattern, Auto Layout, Core Data, MapKit, APNs, Keychain, URLSession, XCTest/XCTest, Instruments, accessibility/WCAG, SwiftLint, Xcode.

AI / On-Device: WebLLM, WebGPU, MLC-AI, local LLMs, Model Context Protocol (MCP), RAG, embeddings, Ollama, prompt/context engineering, model benchmarking, Claude Code, Cursor.

Web / Cloud: React, TypeScript, Node.js, Chrome Extension MV3, REST/GraphQL, AWS (Lambda, DynamoDB, S3), Supabase, PostgreSQL, Python, Docker, Jenkins, GitHub Actions.

PROFESSIONAL EXPERIENCE

Cyan Technologies

iOS Developer

Chicago, IL / Remote
Nov 2025 - Jul 2026

- Led migration of a **45K-line legacy UIKit codebase to SwiftUI**, completing the transition three weeks ahead of schedule.
- Cut **app launch time by 55%** through lazy loading and memory-footprint optimization, measured in Instruments.
- Automated **~80% of regression testing** with XCTest, shrinking the manual QA cycle and catching regressions pre-release.

Walmart Inc. (via Apolis Inc. & Geo Tech Corp.)

iOS Developer III

Sunnyvale, CA / Hybrid
May 2024 - Aug 2025

- Single-handedly drove the **org-wide app-size-reduction initiative** for the post-transactions team, cutting its three owned modules by **~50%**; traced part of the bloat to an Xcode build behavior expanding small PDF/JPEG assets into multiple PNG renditions, and handed the platform team a remediation plan projected to save **2-3 MB of total app size**.
- Built the new **guest check-in flow end to end**: authored its GraphQL API (adapting legacy REST patterns), drove the API-contract discussions, and shipped the guest-facing UI with deep-link navigation into the new screens.
- Redesigned deep-linking and push-notification navigation using the Coordinator Pattern, enabling reliable routing into nested product views across the flagship US shopping app (**100M+ downloads**).
- Optimized UICollectionView rendering and network data layers with Swift Concurrency, reducing main-thread blocking by 40% on profiled flows (Instruments-verified); integrated Combine-based analytics contributing to a 15% adoption lift.
- Beyond assigned scope: built an **LLM-powered diagnostics dashboard** with the backend orchestration team to triage post-transaction failures, and prototyped **Siri voice access** to the orders and check-in flows at an internal hackathon.

EITACIES Inc.

Software Developer

Chicago, IL / Remote
Jun 2023 - Apr 2024

- Architected HIPAA-compliant healthcare mobile features (Swift, UIKit, MVVM, Core Data, APNs, Keychain) with secure authentication, local persistence, and push-notification workflows.
- Modernized REST networking from callbacks to structured concurrency with resilient retry logic for unreliable hospital networks; led SwiftLint and code-review practices adopted across the mobile team.

University of Illinois Chicago

Full Stack Developer / Graduate Assistant

Chicago, IL
Aug 2022 - May 2023

- Engineered secure Django/React internal dashboards (2FA, RBAC, SQL-backed workflows) and automated administrative processes end-to-end, cutting processing time for department staff.

Humourbaba Pvt. Ltd.

Software Developer

India
Dec 2019 - Jan 2021

- Re-architected the core web platform on Node.js and MongoDB (500+ daily transactions) with JWT-based stateless authentication.

OPEN SOURCE, AI INFRASTRUCTURE & PRODUCT

Gemma 4 WebGPU Engine - Author (clean-room)

Aug 2026 - Present

- Wrote a browser-native inference engine for **Gemma 4 E2B** from Google's published architecture and the transformers reference, with **no code from any third-party engine**; every matmul is a hand-written WGSL compute shader, weights stream from safetensors into IndexedDB. Apache-2.0.
- Cuts **59.0% off every prefill chunk, bit-identically**, by running only the 15 of 35 layers that produce KV cache; **1.73x-2.33x faster time-to-first-token** than the engine it replaced (interleaved A/B, one machine, eight prompts). Splits a **1.17 GB** per-layer embedding table across vocabulary rows so the model runs inside iOS's `maxBufferSize` ceiling. github.com/Ar5en1c/gemma4-webgpu-engine

NVIDIA/D3D12 Subgroup Mismatch - Root Cause & Fix

Jul 2026

- Diagnosed silent output corruption hitting every NVIDIA/Windows user of a widely used WebGPU LLM engine: diffed all **59 compiled kernel pairs on-device**, isolated 6 divergent subgroup reductions, and reduced it to a **60-line engine-free reproducer failing 20/20 trials**. Shipped a fix gated by an on-device self-test rather than vendor detection, restoring **213-223 tok/s** vs 193 for the community workaround. Open Chromium bug against Tint's HLSL backend. github.com/Ar5en1c/gemma4-webgpu-nvidia-subgroup-fix · crbug.com/535116173

YouxAI - Founder / Lead Engineer

Aug 2025 - Present

- Built a privacy-first job-search copilot (youxai.app): Chrome extension + web platform with browser-local LLM inference (WebGPU, Gemma 4 E2B on-device), autofill across Ashby, Workday, Greenhouse, Lever and LinkedIn, and a pipeline indexing **94k+ jobs refreshed daily**.
- Grew to **263 signups** organically on a ~\$25/month backend; model inference runs on-device, so resumes never go to a model provider. Grew out of a 2023 Chrome-extension tracker (React 18, AWS Lambda, DynamoDB) that held organic weekly actives while unmaintained.

WebLLM Bench - Creator

Mar 2026

- Open-sourced a browser-native WebGPU LLM benchmark with reproducible JSON reports (latency, decode throughput, long-context behavior); custom-compiled and **published an 8192-context Qwen2.5-1.5B MLC model to Hugging Face** (huggingface.co/Ar5en1c).

Context Guardian - Creator

Apr 2026 - Present

- Built an npm-published MCP server and transparent LLM proxy ([npm: context-guardian-mcp](https://npm:context-guardian-mcp)) that indexes prompts locally (Ollama/nomic-embed) so cloud models retrieve precisely instead of ingesting raw context; benchmarked **~24% token reduction** across three scenarios, with one regressing, which is what the harness exists to catch.

Headroom - Creator

Jul 2026

- Built a 30-second in-browser measure of a device's real memory-bandwidth ceiling, the number that actually bounds batch-1 decode, using **three independent probes and a from-scratch GEMV ladder**; ships an on-device driver canary for the mismatch above. github.com/Ar5en1c/headroom

PUBLICATIONS & COMMUNITY

- Technical deep-dive on the NVIDIA/Windows WebGPU mismatch (ar5en1c.hashnode.dev); further articles on local LLMs, MCP, and agentic AI. Developer posts with 120k+ LinkedIn and 160k+ Reddit views.

EDUCATION

Indiana Wesleyan University - MS, Computer Information Systems (AI Specialization)

Aug 2026 - May 2028

University of Illinois Chicago - MS, Computer Science

Aug 2021 - May 2023

Vellore Institute of Technology - BTech, Computer Science

Jul 2017 - May 2021